

BAYES ESTIMATORS OF THE LIFETIME PARAMETERS USING THE COMPOUND RAYLEIGH MODEL

P.J. Mostert, J.J.J. Roux & A. Bekker**

Department of Statistics and Actuarial Science, University of Stellenbosch,
Private Bag X1, Matieland, 7602, South Africa

*Department of Statistics, University of South Africa, P. O. Box 392, Pretoria,
0003, South Africa

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Summary: The compound Rayleigh distribution, with its unimodal hazard function, makes it attractive for modelling lifetimes of patients with characteristics of a random hazard rate. The Bayes estimators for some lifetime parameters, as well as the parameters of the compound Rayleigh model, are derived for a right censored sample. The estimators for these parameters are obtained, using the squared error loss function, Varian's linear-exponential (linex) loss function and a weighted linex loss function. A discrete prior probability distribution is placed on the scale parameter and either the noninformative or the conjugate prior distribution is placed on the shape parameter. The prediction of a future lifetime is derived, using the noninformative and the conjugate prior distributions. An example illustrates the proposed estimators for the compound Rayleigh model.

1. The compound Rayleigh model

Bhattacharya and Tyagi (1990) used the Rayleigh distribution to model the survival time distribution of cancer patients in certain clinical studies. The probability density function (pdf) of this distribution is

$$f(t; \theta) = 2\theta t e^{-\theta t^2},$$

where $t > 0$ is the lifetime, and $\theta > 0$. The corresponding hazard function is

$$h(t) = 2\theta t, \quad t > 0.$$

If the patients are sampled from populations with variable hazard functions, i.e. if θ is itself a random variable, then the distribution of lifetimes is a compound Rayleigh distribution. The particular form of θ , which is considered here, is the gamma pdf

$$g(\theta; \beta, \gamma) = \frac{\beta^\gamma \theta^{\gamma-1} e^{-\beta\theta}}{\Gamma(\gamma)}, \quad \theta, \beta, \gamma > 0.$$

The parameters β and γ are called the scale and shape parameters, respectively. The resulting compound distribution has pdf

$$f(t; \beta, \gamma) = 2\gamma\beta^\gamma t(\beta + t^2)^{-(\gamma+1)},$$

which is referred to as the compound Rayleigh distribution.

The lifetime parameters, namely the hazard function, the mean survival time and the survival distribution function, are estimated in this paper. These parameters are of great interest in the analysis of survival data (Leemis, 1986). The hazard function of this compound Rayleigh distribution is unimodal in shape and is given by

$$h(t) = \frac{2\gamma t}{\beta + t^2}. \quad (1)$$

This unimodal hazard function (1) is particularly useful when the time of the peak failure rate is of prime interest (Greenwich, 1992).

The hazard function (1) has one critical point at

$$t = \sqrt{\beta} \quad (2)$$

and is also the maximum point. The height of $h(t)$ is the maximum value of (1), i.e.

$$h(\sqrt{\beta}) = \frac{\gamma}{\sqrt{\beta}}.$$

Thus, for the peak time $t = \sqrt{\beta}$, the height of $h(t)$ can be controlled by the parameter γ (see figure 1 (curves A and B)). Figure 1 gives the plot of $h(t)$ (1) for different values of β and γ .

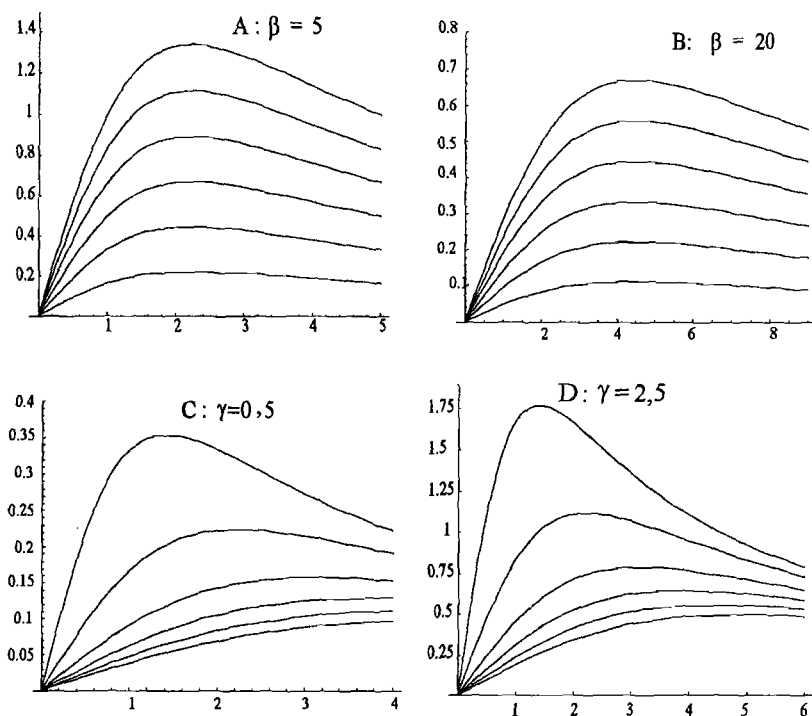


Figure 1. Hazard function of the compound Rayleigh model
A and B: γ -values 0,5; 1,0; 1,5; 2,0; 2,5; 3,0 from bottom to top
C and D: β -values 2; 5; 10; 15; 20; 25 from top to bottom

Examination of figure 1 (curves C and D), shows that for a fixed γ , the maximum point of $h(t)$ shifts to the right, for an increasing scale parameter β .

Since $h(t)$ is a decreasing function for large values of t , the compound Rayleigh model seems implausible as lifetime model in most situations. Nevertheless, the compound Rayleigh model is often found suitable for representing lifetimes, especially when very large values of t are not of interest (Abdel-Ghaly & Attia, 1993).

The survival distribution function and the mean survival time of the compound Rayleigh model are given by

$$S(t) = \beta^\gamma (\beta + t^2)^{-\gamma}, \quad t > 0 \quad (3)$$

and

$$\mu = \frac{\sqrt{\beta\pi}\Gamma(\gamma - \frac{1}{2})}{2\Gamma(\gamma)}, \quad (4)$$

respectively.

Remark 1 The first moment (4) of the compound Rayleigh model is only defined for γ -values larger than 0,5. The second moment, i.e. the variance, is only defined for γ -values larger than 1,0.

In the light of this remark, it may be necessary to use other measures of location instead of μ for the compound Rayleigh model, for example the mode

$$mod = \sqrt{\frac{\beta}{2\gamma + 1}} \quad (5)$$

and the median

$$med = \sqrt{\beta \left(2^{\frac{1}{\gamma}} - 1 \right)}. \quad (6)$$

2. Bayes estimators for the lifetime parameters

2.1 Likelihood function

Suppose that a right censored sample consists of the n patients, where only d lifetimes, $t_1 < t_2 < \dots < t_d$, are fully measured, while the remainder $n - d$ are censored. These $n - d$ censored lifetimes will be ordered separately and are denoted by $t_{d+1} < t_{d+2} < \dots < t_n$. The likelihood function for a right censored sample (Dellaportas & Wright, 1991) from the compound Rayleigh model, is given by

$$\mathcal{L}(\beta, \gamma) \propto (2\gamma)^d u e^{-\gamma T} \quad (7)$$

where

$$T = \sum_{i=1}^n \ln \left[1 + \frac{t_i^2}{\beta} \right] \quad (8)$$

and

$$u = \prod_{i=1}^d \frac{t_i}{\beta + t_i^2}.$$

Maximum likelihood estimators for β and γ cannot be obtained in closed form expressions, but can be obtained by using numerical methods. The maximum likelihood estimators for the lifetime parameters can then be obtained by substituting the maximum likelihood estimates of β and γ into expressions (1), (3) and (4).

2.2 Prior information

Bayes estimation of the parameters in this paper is done with respect to the noninformative and the conjugate priors. Since T in (8) depends on the parameter β , a continuous joint prior pdf of β and γ will result in a posterior pdf, where the normalising constant will involve integrals which cannot be evaluated explicitly. The method of Soland (1969) results in closed form

expressions for the posterior pdfs. A discrete prior probability distribution is placed on β and a continuous prior distribution on γ of the compound Rayleigh model.

Suppose the parameter space of β is restricted to a finite number of values, say $\beta_1, \beta_2, \dots, \beta_m$ in the interval $(0, \infty)$, with the following probability distribution of β :

$$\zeta_j = \Pr(\beta = \beta_j), \quad \sum_{j=1}^m \zeta_j = 1, \quad \text{for } j = 1, \dots, m.$$

The joint posterior pdf of γ and β is given by

$$g(\gamma, \beta_j | T_j) \propto \mathcal{L}(\gamma, \beta_j) g(\gamma | \beta_j) \Pr(\beta = \beta_j), \quad (9)$$

where

$$\mathcal{L}(\gamma, \beta_j) \propto (2\gamma)^d u_j e^{-\gamma T_j},$$

with

$$T_j = \sum_{i=1}^n \ln \left[1 + \frac{t_i^2}{\beta_j} \right] \quad (10)$$

and

$$u_j = \prod_{i=1}^d \frac{t_i}{\beta_j + t_i^2}. \quad (11)$$

The continuous prior pdf, $g(\gamma | \beta_j)$, is either taken to be the noninformative or the conjugate prior. These are discussed in the subsequent sections.

2.2.1 Noninformative prior

The approximate noninformative prior pdf of γ given $\beta = \beta_j$, using Jeffreys' rule (Box & Tiao, 1992), is obtained as

$$g(\gamma | \beta_j) \propto \frac{1}{\gamma}, \quad \gamma > 0. \quad (12)$$

The posterior pdf of γ given $\beta = \beta_j$, using (9) and (12), is given by

$$g(\gamma | T_j; \beta_j) \propto \gamma^d e^{-\gamma T_j} \frac{1}{\gamma}.$$

Using the gamma function integral

$$\int_0^{\infty} x^{v-1} e^{-\rho x} dx = \Gamma(v) \rho^{-v}, \quad v > 0, \quad (13)$$

the posterior pdf becomes

$$g(\gamma | T_j; \beta_j) = \frac{T_j^d \gamma^{d-1} e^{-\gamma T_j}}{\Gamma(d)}, \quad (14)$$

which is the pdf of a gamma distribution with parameters T_j and d .

The marginal posterior probability distribution of β , using (9) and (12), is obtained as

$$p_j \equiv \Pr(\beta = \beta_j | T_j) \propto \int_0^{\infty} \gamma^d u_j e^{-\gamma T_j} \frac{1}{\gamma} \zeta_j d\gamma.$$

Using (13), this reduces to

$$p_j = \frac{\zeta_j u_j \Gamma(d) T_j^{-d}}{\sum_{j=1}^m \zeta_j u_j \Gamma(d) T_j^{-d}}, \quad \text{for } j = 1, \dots, m.$$

2.2.2 Conjugate prior

The conjugate prior pdf of γ given $\beta = \beta_j$, is taken to be the gamma pdf with parameters b_j and r_j and depends on β_j only through the hyperparameters b_j and r_j , i.e.

$$g(\gamma | \beta_j) = \frac{b_j^{r_j} \gamma^{r_j-1} e^{-\gamma b_j}}{\Gamma(r_j)}, \quad \gamma, r_j, b_j > 0. \quad (15)$$

The posterior pdf of γ given $\beta = \beta_j$, using (9) and (15), is given by

$$g(\gamma | T_j; \beta_j) \propto \gamma^d e^{-\gamma T_j} \gamma^{r_j-1} e^{-\gamma b_j}.$$

Using (13), the posterior pdf becomes

$$g(\gamma | T_j; \beta_j) = \frac{(T_j + b_j)^{r_j+d} \gamma^{r_j+d-1} e^{-\gamma(T_j+b_j)}}{\Gamma(r_j+d)}, \quad (16)$$

which is the pdf of a gamma distribution with parameters $T_j + b_j$ and $d + r_j$.

The marginal posterior probability distribution of β , using (9) and (15), is obtained as

$$p_j \equiv \Pr(\beta = \beta_j | T_j) \propto \int_0^\infty \gamma^d u_j e^{-\gamma T_j} \frac{b_j^{r_j} \gamma^{r_j-1} e^{-\gamma b_j}}{\Gamma(r_j)} \zeta_j d\gamma.$$

Using (13), this reduces to

$$p_j = \frac{\zeta_j u_j b_j^{r_j} \Gamma(r_j+d) (T_j + b_j)^{-r_j-d} [\Gamma(r_j)]^{-1}}{\sum_{j=1}^m \zeta_j u_j b_j^{r_j} \Gamma(r_j+d) (T_j + b_j)^{-r_j-d} [\Gamma(r_j)]^{-1}} \quad \text{for } j = 1, \dots, m. \quad (17)$$

It is, however, not always possible to know the values of the hyperparameters in (15). A method for obtaining values for b_j and r_j is to use the expected value of $S(t)$ (from (3)) conditional on β_j , which is given by

$$E_{\gamma|\beta_j} \left[\beta_j^\gamma (\beta_j + t^2)^{-\gamma} \right] = \int_0^\infty e^{-\gamma \ln \left[1 + \frac{t^2}{\beta_j} \right]} \frac{b_j^{r_j} \gamma^{r_j-1} e^{-\gamma b_j}}{\Gamma(r_j)} d\gamma, \quad j = 1, \dots, m,$$

since the conditional prior of γ given $\beta = \beta_j$, is gamma distributed. Using (13), this simplifies to

$$E_{\gamma|\beta_j} \left[\beta_j^\gamma (\beta_j + t^2)^{-\gamma} \right] = \left(\frac{\ln \left[1 + \frac{t^2}{\beta_j} \right] + b_j}{b_j} \right)^{-r_j}, \quad j = 1, \dots, m. \quad (18)$$

Thus, for two prior probabilities of cancer patients to survive t , the values of b_j and r_j for each β_j , can be obtained numerically from (18).

Remark 2 The posterior pdfs, when using the noninformative and conjugate priors respectively, are both pdfs of the gamma distribution (see (14) and (16)). Hence, the derivation of the estimators will be similar

Mostert, Bekker and Roux (1998) obtained Bayes estimators for the lifetime parameters under the squared error loss function and the linex loss function using the Rayleigh model. The estimation of the lifetime parameters and the parameters under the squared error loss function and linex loss function is discussed in the subsequent sections for the conjugate prior and summarised in tables 1 and 2 for the noninformative prior

2.3 Under the squared error loss function

The Bayes estimator for γ is

$$\hat{\gamma}_S = \int_0^{\infty} \sum_{j=1}^m \gamma g(\gamma | T_j; \beta_j) d\gamma.$$

Using (13) and (16) this simplifies to

$$\hat{\gamma}_S = \sum_{j=1}^m \frac{p_j(r_j + d)}{T_j + b_j}. \quad (19)$$

From (1), the Bayes estimator for $h(t)$ is

$$\hat{h}_S(t) = 2t \sum_{j=1}^m \frac{p_j(r_j + d)}{(\beta_j + t^2)(T_j + b_j)}. \quad (20)$$

Similarly from (3), the Bayes estimator for $S(t)$ follows as

$$\hat{S}_S(t) = \sum_{j=1}^m p_j \left(1 + \frac{\ln[1 + \frac{t^2}{\beta_j}]}{T_j + b_j} \right)^{-(r_j + d)}. \quad (21)$$

The Bayes estimators for μ and the median cannot be obtained in closed form. As an alternative, $\hat{\gamma}_S$ can be substituted into (4) and (6) to obtain Bayes estimators for μ and the median, respectively. Restrictions, however, exist in the case of μ (see remark 1).

From (5) the estimator for the mode is

$$\widehat{mod}_S = \sum_{j=1}^m p_j \frac{(T_j + b_j)^{r_j+d} \sqrt{\beta_j}}{\Gamma(r_j + d)} \int_0^{\infty} (2\gamma + 1)^{-\frac{1}{2}} \gamma^{r_j+d-1} e^{-\gamma(T_j+b_j)} d\gamma.$$

Using the confluent hypergeometric function integral

$$\int_0^{\infty} \frac{x^{v-1} e^{-\kappa x}}{(1+x)^{1+v-\eta}} dx = \Gamma(v) \Psi(v, \eta, \kappa), \quad v > 0, \quad (22)$$

(Gradshteyn & Ryzhik, 1965)

this reduces to

$$\widehat{mod}_S = \sum_{j=1}^m p_j \sqrt{\beta_j} \left(\frac{T_j + b_j}{2} \right)^{r_j+d} \Psi \left(r_j + d, r_j + d + \frac{1}{2}; \frac{T_j + b_j}{2} \right). \quad (23)$$

The Bayes estimators for the parameters, in the case of the noninformative prior, are obtained from the Bayes estimators when using the conjugate prior, by substituting the parameters T_j and d of the gamma posterior pdf (14) into the corresponding parameters $T_j + b_j$ and $d + r_j$ of the posterior pdf (16) and are summarised in table 1.

Table 1. Bayes estimators under the squared error loss function

Parameter	Noninformative prior	Conjugate prior
γ	$\sum_{j=1}^m p_j \frac{d}{T_j}$	(19)
$h(t)$	$2t \sum_{j=1}^m p_j \frac{d}{(\beta_j + t^2)T_j}$	(20)
$S(t)$	$\sum_{j=1}^m p_j \left(1 + \frac{\ln[1 + \frac{t^2}{\beta_j}]}{T_j} \right)^{-d}$	(21)
mod	$\sum_{j=1}^m p_j \sqrt{\beta_j} \left(\frac{T_j}{2} \right)^d \Psi \left(d, d + \frac{1}{2}; \frac{T_j}{2} \right)$	(23)

2.4 Under the linex loss function

The linex loss function is defined as

$$L_L(\Delta) = e^{a\Delta} - a\Delta - 1, \quad a \neq 0, \quad (24)$$

where Δ is the scalar estimation error (Varian, 1975). The parameter a reflects the direction of asymmetry and its magnitude the degree of asymmetry. If one does not know what value of a to choose, a possibility is to construct a weighted linex loss function, i.e. the weighted sum of (24) with different values for the parameter a .

Remark 3 The weighted linex loss function is defined as

$$L_{WL}(\Delta) = w_1 L_{L1}(\Delta) + \dots + w_k L_{Lk}(\Delta),$$

where $\sum_{i=1}^k w_i = 1$ and $L_{Li}(\Delta)$ represents the linex loss function (24) with $a = a_i$, $i = 1, \dots, k$.

The Bayes estimator for γ , under the linex loss function (24), is

$$\hat{\gamma}_L = -\frac{1}{a} \ln \left[\int_0^\infty \sum_{j=1}^m p_j e^{-a\gamma} g(\gamma | T_j; \beta_j) d\gamma \right] \quad (25)$$

(Zellner, 1986).

Using (13) this simplifies to

$$\hat{\gamma}_L = -\frac{1}{a} \ln \left[\sum_{j=1}^m p_j \left(1 + \frac{a}{T_j + b_j} \right)^{-(r_j+d)} \right]. \quad (26)$$

Similarly from (1) and (13) the Bayes estimator for $h(t)$, follows as

$$\hat{h}_L(t) = -\frac{1}{a} \ln \left[\sum_{j=1}^m p_j \left(1 + \frac{2at}{(\beta_j + t^2)(T_j + b_j)} \right)^{-(r_j+d)} \right]. \quad (27)$$

The Bayes estimator for $S(t)$ is

$$\hat{S}_L(t) = -\frac{1}{a} \ln \left[\int_0^\infty \sum_{j=1}^m p_j e^{-a\left(\frac{\beta_j}{\beta_j+t^2}\right)\gamma} \frac{(T_j+b_j)^{r_j+d} \gamma^{r_j+d-1} e^{-\gamma(T_j+b_j)}}{\Gamma(r_j+d)} d\gamma \right].$$

Using the exponential series

$$e^v = \sum_{k=0}^{\infty} \frac{v^k}{k!}, \quad -\infty < v < \infty$$

and (13), this simplifies to

$$\hat{S}_L(t) = -\frac{1}{a} \ln \left[\sum_{k=0}^{\infty} \sum_{j=1}^m \frac{(-a)^k p_j}{k!} \left(1 + \frac{k \ln \left[1 + \frac{t^2}{\beta_j} \right]}{T_j + b_j} \right)^{-(r_j+d)} \right]. \quad (28)$$

The Bayes estimators for μ and the median cannot be obtained in closed form. As an alternative, $\hat{\gamma}_L$ can be substituted into (4) and (6) to obtain Bayes estimators for μ and the median, respectively.

The Bayes estimator for the mode, using the exponential series, is

$$\widehat{mod}_L = -\frac{1}{a} \ln \left[\int_0^\infty \sum_{j=1}^m p_j e^{-a\sqrt{\frac{\beta_j}{2\gamma+1}}(T_j+b_j)} \frac{\gamma^{r_j+d-1} e^{-\gamma(T_j+b_j)}}{\Gamma(r_j+d)} \right]$$

Using (22), this simplifies to

$$\widehat{mod}_L = -\frac{1}{a} \ln \left[\sum_{k=0}^\infty \sum_{j=1}^m p_j \frac{(-a)^k (\beta_j)^{\frac{k}{2}}}{k!} \left(\frac{T_j+b_j}{2} \right)^{r_j+d} \times \Psi \left(r_j+d, r_j+d+1-k; \frac{T_j+b_j}{2} \right) \right]. \quad (29)$$

Table 2 summarises the Bayes estimators for the parameters using the noninformative prior

Table 2. Bayes estimators under the linex loss function

Parameter	Noninformative prior	Conjugate prior
γ	$-\frac{1}{a} \ln \left[\sum_{j=1}^m p_j \left(1 + \frac{a}{T_j} \right)^{-d} \right]$	(26)
$h(t)$	$-\frac{1}{a} \ln \left[\sum_{j=1}^m p_j \left(1 + \frac{2at}{(\beta_j + t^2)T_j} \right)^{-d} \right]$	(27)
$S(t)$	$-\frac{1}{a} \ln \left[\sum_{k=0}^\infty \sum_{j=1}^m p_j \frac{(-a)^k}{k!} \right]$	(28)
	$\left(1 + \frac{k \ln[1 + \frac{t^2}{\beta_j}]}{T_j} \right)^{-d}$	
mod	$-\frac{1}{a} \ln \left[\sum_{k=0}^\infty \sum_{j=1}^m p_j \frac{(-a)^k (\beta_j)^{\frac{k}{2}}}{k!} \left(\frac{T_j}{2} \right)^d \right]$	(29)
	$\Psi \left(d, d+1-k; \frac{T_j}{2} \right)$	

3. Prediction of a future survival time

The prediction considered in this paper is that of a future patient's lifetime, say, $y > t_d$, i.e. the next lifetime from the sample (Howlader & Hossian, 1995).

Let $z = y - t_d$. The posterior pdf of z , using the conjugate prior, is given by

$$g(z | T_j; \beta_j) = \sum_{j=1}^m \frac{2p_j z (T_j + b_j)^{r_j+d} z}{(\beta_j + z^2) \Gamma(r_j + d)} \int_0^\infty e^{\gamma \ln \left[\frac{\beta_j}{\beta_j + z^2} \right]} \gamma^{r_j+d} e^{-\gamma(T_j+b_j)} d\gamma.$$

Using (13), this simplifies to

$$g(z | T_j; \beta_j) = \sum_{j=1}^m \frac{2z p_j (r_j + d) (T_j + b_j)^{r_j+d}}{(\beta_j + z^2) \left[T_j + b_j - \ln \left(\frac{\beta_j}{\beta_j + z^2} \right) \right]^{r_j+d+1}}. \quad (30)$$

From (30) the survival distribution function of z simplifies to

$$\Pr(Z > t | T_j; \beta_j) = \sum_{j=1}^m p_j \left(\frac{T_j + b_j}{T_j + b_j - \ln \left[\frac{\beta_j}{\beta_j + t^2} \right]} \right)^{r_j+d}.$$

The $100(1 - \alpha)\%$ lower and upper prediction bounds of the credible region (R) of z are solutions of t in

$$\sum_{j=1}^m p_j \left(\frac{T_j + b_j}{T_j + b_j - \ln \left[\frac{\beta_j}{\beta_j + t^2} \right]} \right)^{r_j+d} = \rho, \quad (31)$$

with $\rho = \alpha$ and $\rho = 1 - \alpha$, respectively.

The posterior moments, using (30), under the squared error loss function and the linex loss function, cannot be obtained in closed form. A possibility for obtaining $\hat{z}_S$ and $\hat{z}_L$, is to use numerical methods such as the Monte Carlo simulation procedure (Kloek & Van Dijk, 1978). This will be illustrated

in section 4. Table 3 summarises the prediction of a future survival time using the noninformative prior.

Table 3. Prediction of a future survival time

Expression	Noninformative prior	Conjugate prior
$g(z T_j; \beta_j)$	$\sum_{j=1}^m \frac{p_j 2z d(T_j + b_j)^d}{(\beta_j + z^2) \left[T_j - \ln \left(\frac{\beta_j}{\beta_j + z^2} \right) \right]^{d+1}}$	(30)
R	$\sum_{j=1}^m p_j \left(\frac{T_j}{T_j - \ln \left[\frac{\beta_j}{\beta_j + t^2} \right]} \right)^d = \rho$	(31)

4. Illustrative example

To illustrate the application of the results to the analysis of survival data, consider the survival times in years of a group of patients given chemotherapy treatment alone, shown in table 4; this is a subset of data reported by Stablein et al. (1981).

Table 4. Lifetimes in years

Chemo-therapy	0,003	0,173	0,288	0,353	0,499	0,592	0,685	0,718	0,825
	0,825	0,937	0,970	0,975	0,981	1,041	1,044*	1,049	1,049
	1,063	1,079	1,118	1,260	1,340	1,367	1,436	1,449*	1,466
	1,540	1,849	1,852	2,049	2,049	2,132	2,153	2,184	2,589*
	2,616	2,652	3,233*	3,411	3,482	3,499*	3,827*	4,142*	4,162*

* Indicates censored observations

Figure 2 shows the Kaplan-Meier estimator for $S(t)$, and $S(t)$ of the Rayleigh (curve B) and the compound Rayleigh (curve A) models, fitted by the method of maximum likelihood. The maximum likelihood estimators for β and γ are obtained as $\hat{\beta}_{MLE} = 2,276$ and $\hat{\gamma}_{MLE} = 1,063$, respectively

using the Newton-Raphson method. The maximum likelihood estimator for θ in the Rayleigh model, was obtained as $\hat{\theta}_{MLE} \approx 0,2125$.

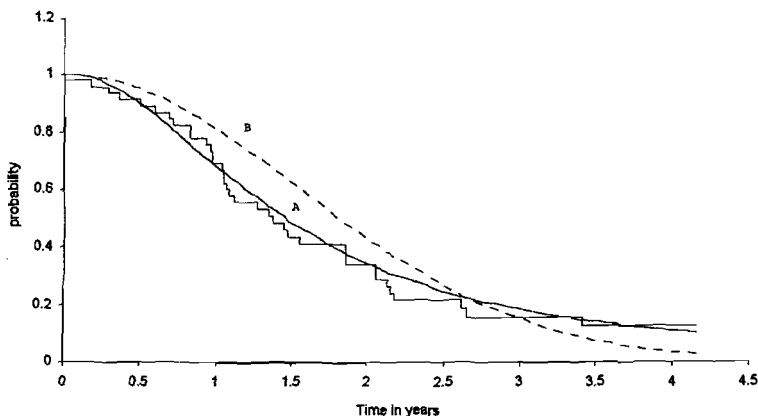


Figure 2. Kaplan-Meier estimator for $S(t)$ and
 A: $S(t)$ of the compound Rayleigh model
 B: $S(t)$ of the Rayleigh model

Remark 4 By substituting $\hat{\beta}_{MLE}$ and $\hat{\gamma}_{MLE}$ into (7), the maximised likelihood function of the compound Rayleigh model is equal to $6,047 \times 10^{-28}$. This can now be compared to the maximised likelihood function of the Rayleigh model, which is equal to $8,167 \times 10^{-31}$. (This Kappenman procedure (Kappenman, 1982) is a method for selecting the best of two models fitted to the data.)

The graph indicates that the compound Rayleigh model is acceptable for these data. This model is suitable for the situation where the medical doctors are interested in the analysis of the shorter lifetimes. They also believe that the cancer patients will have an increasing $h(t)$ for at least $t = 2,0$, after which a decreasing $h(t)$ can be expected. The values assumed for β , with

equal prior probabilities, are therefore chosen to ensure this increasing $h(t)$ for most of the lifetimes (see figure 1 (curves C and D)), by using (2). The prior information concerning β , as well as the values of T_j and $u_j, j = 1, \dots, 4$, using (10) and (11) respectively, is summarised in table 5.

Table 5. Prior information for β of the compound Rayleigh model

j	1	2	3	4
β_j	2,5	3,5	4,5	5,5
ζ_j	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
T_j	32,916	26,683	22,606	19,692
u_j	$6,55 \times 10^{-25}$	$2,06 \times 10^{-28}$	$3,09 \times 10^{-31}$	$1,31 \times 10^{-33}$

The illustration of the results in this paper is done for the conjugate prior on the shape parameter. The hyperparameters of the gamma prior (15) are derived, using (18). From experience, a doctor assumes that the probabilities are respectively 0,24 and 0,45 that a patient will live longer than 2,3 and 1,4 years. These two prior probabilities are substituted into (18), where b_j and r_j are solved numerically for each given $c_j, j = 1, \dots, 4$, using the Newton-Raphson method. Table 6 gives the values of the hyperparameters in the gamma prior, as well as the posterior probabilities (see 17), derived for each β_j in turn.

Table 6. Hyperparameter values of the gamma prior and posterior probabilities

j	1	2	3	4
β_j	2,5	3,5	4,5	5,5
r_j	3,547	2,310	1,860	1,639
b_j	2,334	1,098	0,688	0,496
p_j	0,4628	0,2696	0,1630	0,1046

Figure 3 gives the plot of the estimates of $h(t)$ for the gamma prior, under the squared error loss function, the linex loss function and the weighted linex loss function (see (20) and (27)).

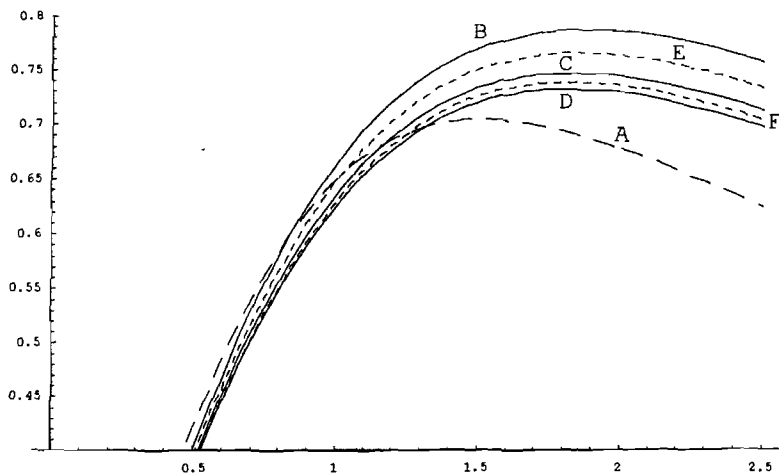


Figure 3. Estimates of the hazard function using the conjugate prior

- A:** $\hat{h}_{MLE}(t)$ **B:** $\hat{h}_L(t); a = -5, 0$ **C:** $\hat{h}_S(t)$
D: $\hat{h}_L(t); a = 2, 0$ **E:** $\hat{h}_{WL}(t); -0, 5; -1, 5; -5, 0$
F: $\hat{h}_{WL}(t); 0, 5; 1, 0; 2, 0$

The prior information regarding β has shifted the peak time of $\hat{h}_S(t)$, $\hat{h}_L(t)$ and $\hat{h}_{WL}(t)$ to the right of $\hat{h}_{MLE}(t)$, resulting in increasing values of the estimates of $h(t)$ for a longer period of time. It is also evident that little differences exist in the shape of $\hat{h}_S(t)$, $\hat{h}_L(t)$ and $\hat{h}_{WL}(t)$. Hence, for a remarkable difference in the estimates of $h(t)$ under the linex loss function, the value of a should be increased or decreased significantly for underestimation or overestimation, respectively.

The Bayes estimates of $\gamma, \mu, \text{mod}, \text{med}$ and $S(t)$ (see tables 1 and 2) are given in table 7.

Table 7. Estimates of the parameters using the conjugate prior

	t	MLE	L_S	L_L						L_{WL}	
				a						a	
				0,5	1,0	2,0	-0,5	-1,5	-5,0	0,5;1,0;2,0	-0,5;-1,5;-5,0
γ		1,063	1,386	1,358	1,332	1,287	1,417	1,488	1,823	1,326	1,576
μ		2,185	1,990	1,964	1,952	1,960	2,028	2,083	2,103	1,959	2,071
mod		0,853	0,951	0,926	0,909	0,892	0,982	1,037	1,043	0,909	1,021
med		1,447	1,487	1,457	1,438	1,425	1,527	1,593	1,614	1,440	1,578
$S(t)$	0,25	0,972	0,974	0,974	0,974	0,974	0,974	0,974	0,975	0,974	0,974
	0,50	0,895	0,904	0,904	0,904	0,904	0,904	0,905	0,905	0,904	0,905
	0,75	0,791	0,806	0,806	0,806	0,805	0,806	0,807	0,808	0,806	0,807
	1,00	0,679	0,697	0,696	0,696	0,695	0,697	0,698	0,702	0,696	0,699
	1,25	0,574	0,591	0,590	0,589	0,588	0,591	0,593	0,597	0,589	0,594
	1,50	0,482	0,495	0,494	0,493	0,492	0,496	0,497	0,503	0,493	0,499
	1,75	0,404	0,413	0,412	0,411	0,409	0,414	0,415	0,421	0,411	0,417
	2,00	0,340	0,344	0,343	0,342	0,341	0,345	0,347	0,353	0,342	0,348
	2,25	0,288	0,288	0,287	0,286	0,284	0,288	0,290	0,296	0,286	0,291
	2,50	0,246	0,242	0,241	0,240	0,239	0,242	0,244	0,249	0,240	0,245
	2,75	0,211	0,204	0,203	0,203	0,201	0,205	0,206	0,211	0,202	0,207
	3,00	0,183	0,174	0,173	0,172	0,171	0,174	0,175	0,180	0,172	0,176

From table 7, the estimates of $S(t)$ under the squared error loss function and linex loss function (for $a = 0,5; 1,0, 2,0 - 0,5$) are smaller than $\hat{S}_{MLE}(t)$ if $t > 2,0$, while $\hat{S}_L(t)$, for $a = -1,5$ and $-5,0$ are smaller than $\hat{S}_{MLE}(t)$, if $t > 2,25$ and $t > 2,5$, respectively.

The estimator for z under the squared error loss function and the linex loss function cannot be obtained in closed form and a Monte Carlo simulation procedure is followed. A sample of 10 000 was generated from the posterior pdf (see (30)), using the Rayleigh model, with $\theta = 0,9$, as importance density. The Bayes predictions of z are summarised in table 8 and the credible regions for z in table 9.

Table 8. Estimates of z using the conjugate prior

a	L_S	L_L						L_{WL}	
		0,5	1,0	2,0	-0,5	-1,5	-5,0	0,5; 1,0; 2,0	-0,5; -1,5; -5,0
z	1,676	2,122	1,727	1,322	1,884	2,485	3,022	1,724	2,464

Table 9. Credible regions for z using the conjugate prior

α	lower	upper
0,90	0,511	3,947
0,95	0,354	5,412
0,99	0,155	11,03

5. Concluding remark

By compounding the Rayleigh model, a two-parameter lifetime model is obtained, which results in greater flexibility in the analysis of cancer lifetimes. The hazard function of the Rayleigh model is monotonically increasing, while that of the compound Rayleigh model has a zero asymptote as t goes to infinity.

The use of a discrete distribution for the shape parameter of the compound Rayleigh model, resulted in a closed form expression for the posterior pdf.

The choice of these values in the discrete distribution for the shape parameter contributed to the usefulness of the model (eg. it increases the peak time of $h(t)$).

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